

Simulating Random Graph Models in Python

Dissertation Presentation

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Why Study Random Graph Models?

Networks are everywhere:

- ▶ Social networks, the internet, neural networks, citation graphs
- ▶ All share structural properties that deterministic models cannot capture

Key structural questions:

- ▶ How do large networks form and evolve?
- ▶ Why do real-world networks have short average path lengths (small-world property)?
- ▶ Why do a small number of nodes attract most connections (hubs)?

Erdős–Rényi

Random edge probability

Watts–Strogatz

Small-world rewiring

Barabási–Albert

Preferential attachment

Goal: Understand the mathematical foundations of three principal random graph models, test theoretical predictions through Python simulation with hundreds of trials per parameter value, and validate against known results.

Three Research Questions

- ▶ **RQ1 — Phase Transition (ER):** At what value of p does the giant component exceed 50% in $G(1000, p)$, and how does this compare with $p_c = 1/n$?
- ▶ **RQ2 — Small-World Window (WS):** Over what range of β does Watts-Strogatz simultaneously maintain high clustering and short average path length?
- ▶ **RQ3 — Power-Law Scaling (BA):** Does the degree distribution follow a power law with $\tau \approx 3$, and does $\langle k \rangle = 2m$ hold across values of m ?

Approach: Theory describes the limit $n \rightarrow \infty$. Simulation shows what actually happens at $n = 1000$.

The Erdős–Rényi Model $G(n, p)$

Definition: Vertices are connected independently with probability p .

Key Metrics:

- ▶ Expected Edges: $p\binom{n}{2}$
- ▶ Avg. Degree: $p(n-1)$
- ▶ Threshold: $p_c = \ln(n)/n$

The Poisson Limit:

In sparse graphs ($p = \lambda/n$), degrees follow a Poisson distribution. *Result: No heavy tails or hubs.*

Erdős–Rényi: Phase Transitions

Subcritical ($\lambda < 1$)

- ▶ Small components
- ▶ Sizes: $O(\log n)$

Critical ($\lambda = 1$)

- ▶ Giant emerges
- ▶ Size: $O(n^{2/3})$

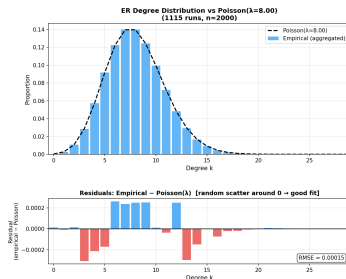
Supercritical ($\lambda > 1$)

- ▶ Unique giant
- ▶ Linear size: $O(n)$

Real-World Gap: $G(n, p)$ misses the **high clustering** and **heavy tails** seen in actual social or web networks.

ER Simulation: Degree Distribution Confirms Poisson

1115 runs · $n = 2000$ · $\lambda = 8.00$ · $p = \lambda/n = 0.004$



1115

simulation runs

RMSE

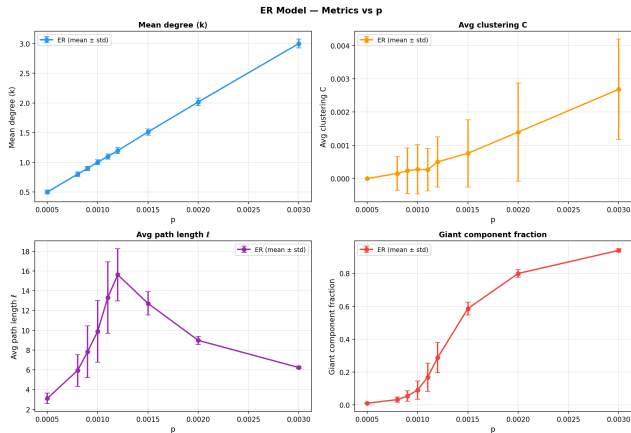
= 0.00015

Near-perfect

theory fit

Conclusion: Empirical degree distribution matches Poisson(8.00) almost exactly. Residuals show random scatter around zero which confirms the Poisson limit theorem numerically.

ER Phase Transition: Giant Component Emergence (RQ1)



RQ1 Answer: empirical 0.5 threshold at $p \approx 0.0014$ vs theory $p_c = 1/n = 0.001$; the difference is consistent with finite-size effects at $n = 1000$.

The Watts–Strogatz Small-World Model

Construction: Ring lattice of n vertices, each linked to k nearest neighbours. Then rewire each edge independently with probability β .

$$\beta = 0$$

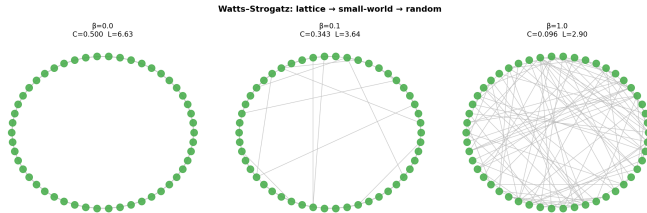
High C , large L .

$$0 < \beta \ll 1$$

Small-world: L drops, C stays high.

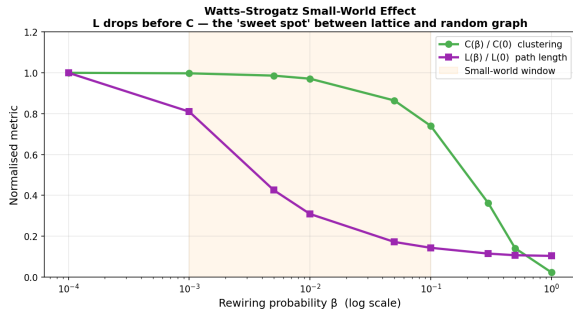
$$\beta = 1$$

Low C , short L .



WS Simulation: The Small-World Window (RQ2)

$n = 500 \cdot k = 8 \cdot \beta$ swept log-uniformly from 10^{-4} to $1 \cdot 30$ runs per point



RQ2 Answer

Small-world window:

$$10^{-3} < \beta < 10^{-1}$$

β	C/C_0	L/L_0
10^{-3}	1.00	0.81
10^{-1}	0.75	0.14

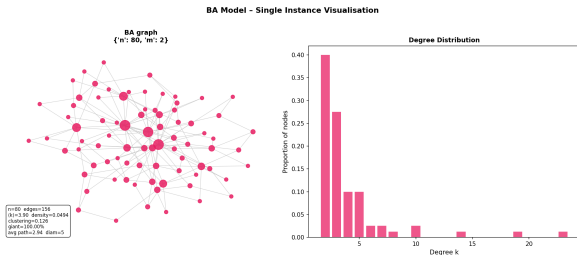
At $\beta = 10^{-3}$: L already falling, C fully intact.

At $\beta = 10^{-1}$: L at 14%, C still 75%.

Key insight: L drops sharply near $\beta \approx 10^{-3}$ because even a few shortcuts bridge distant regions, while C stays high as local lattice structure is preserved.

The Barabási–Albert Preferential Attachment Model

Rule: At each step, add one new vertex with m edges. Connect to existing vertex i with probability $\Pi(i) = d_i / \sum_j d_j$. “The rich get richer.”



Theoretical prediction:

$$P(k) \sim k^{-3}$$

Power law with $\tau = 3$, independent of m .

Hubs are rare — but inevitable.

Mean degree:

$$\langle k \rangle = 2m \quad (\text{edge-counting})$$

Barabási–Albert: Does Simulation Confirm the Theory?

RQ3 asks: Does the simulated degree distribution actually follow $P(k) \sim k^{-3}$, and does $\langle k \rangle = 2m$ hold?

What theory predicts:

- ▶ Slope of -3 in log-log space
- ▶ $\langle k \rangle = 2m$ for all m
- ▶ Hubs with degrees far exceeding the mean

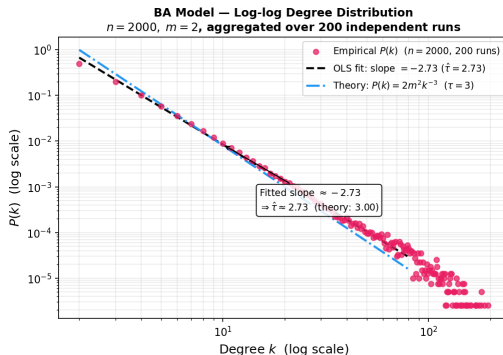
What simulation must show:

- ▶ A linear trend in log-log degree distribution
- ▶ Slope converging toward -3 as n grows
- ▶ Tight $\langle k \rangle = 2m$ relation with low scatter

Theory is asymptotic ($n \rightarrow \infty$). At finite n , the power-law tail truncates — the slope will be pulled above -3 . The question is: by how much, and does it match the theoretical finite-size prediction?

BA Simulation: Power-Law Degree Distribution (RQ3)

200 runs · $n = 2000$ · $m = 2$ · OLS fit over $k \in [2, 50]$ in log-log space



RQ3 Answer

-2.73

OLS fitted slope

($R^2 = 0.998$)

$\tau = 3$

Theory prediction

(asymptotic)

$\langle k \rangle = 2m$

Verified across

$m \in \{1, 2, 3, 5, 8, 12\}$

Slope of $-2.73 < -3$ due to finite- n tail truncation

at $k \approx 50-80$. As $n \rightarrow \infty$ slope converges to -3 .

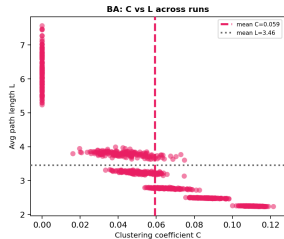
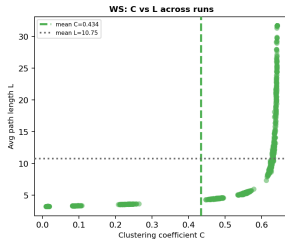
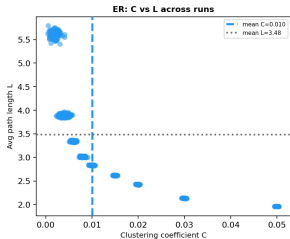
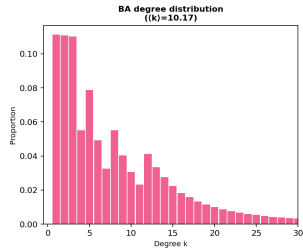
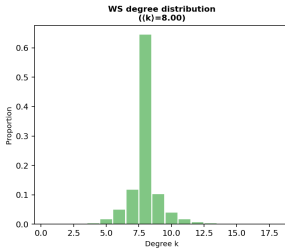
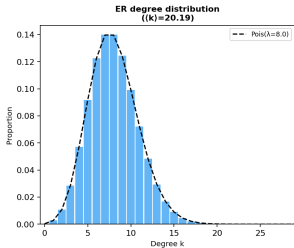
Comparative Structural Properties

	Erdős–Rényi	Watts–Strogatz	Barabási–Albert
Mechanism	Independent edge prob. p	Rewiring a ring lattice with prob. β	Growth + preferential attachment
Degree dist.	Binomial $\rightarrow$ Poisson(λ)	Near-symmetric, narrow spread	Power law $P(k) \sim k^{-3}$
Clustering	Low ($\approx p$)	High for small β	Low–moderate
Hubs	None	None	Present (heavy tail)
Path length	$O(\log n / \log \log n)$	$O(\log n)$ in small-world regime	Short due to hub routing
Mean degree	$\langle k \rangle = (n-1)p$	Exactly preserved at k	$\langle k \rangle = 2m$
Giant component	Threshold at $p_c = 1/n$	Always 1 (edges preserved)	Always 1 (by construction)
Typical uses	Thresholds, percolation theory	Social, neural networks	Web, citation, scale-free nets

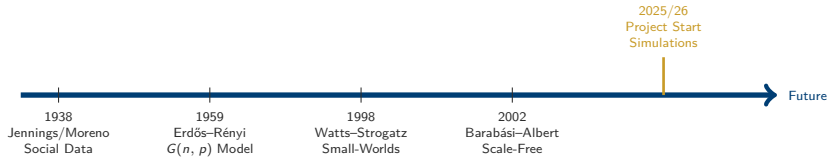
Model Comparison: ER | WS | BA

Top: degree distributions

Bottom: clustering vs path length



The Story of Random Graphs



The field shifted from theoretical probability to modeling the "Small-World" and "Scale-Free" structures of our modern digital life.

Conclusions: Research Questions Answered

- ▶ **RQ1 — Phase Transition:** Threshold at $p \approx 0.0014$ vs. theory $p_c = 0.001$. Discrepancy explained by finite-size effects at $n = 1000$.
- ▶ **RQ2 — Small-World Window:** Confirmed: $10^{-3} < \beta < 10^{-1}$. At lower boundary, L already falling while C remains intact.
- ▶ **RQ3 — Power Law:** Fitted slope -2.73 vs. asymptotic -3 . $\langle k \rangle = 2m$ verified with negligible scatter.

Takeaway: No single model is perfect. Real networks need both local clustering (WS) and global hubs (BA) — and ER shows us where the phase transition lives.